# Arista Access Node BTM-ACC-19: Optical Transport Failure Resolution SOP

**Node:** BTM-ACC-19 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_7583

## Description

This Standard Operating Procedure (SOP) outlines the steps necessary to diagnose and resolve an Optical Transport Failure on the Arista Access Node BTM-ACC-19. The failure is characterized by the alarm code ALARM_7583, indicating a minor severity optical transport issue. This can be attributed to the degradation of the SFP/XFP transceivers, fiber cuts, or a reduction in Rx/Tx power loss. The underlying protocol or hardware mechanics responsible for this failure domain can be attributed to the SONET/SDH protocol, which operates at the OSI layer 1. This protocol relies on the optical transport of light signals over fiber optics to transmit data between nodes. When a SFP/XFP transceiver or fiber optic cable degrades or is compromised, the integrity of the light signal is disrupted, resulting in a failure of the optical transport. In the context of the Arista Access Node, this failure is further complicated by the use of a proprietary CLI dialect. As such, this SOP will adhere strictly to the Arista CLI dialect to ensure accurate and effective troubleshooting and resolution.

## Pre-requisite Checks

- Pre-requisite: Verify that the site is accessible and that remote hands are available to assist, if necessary.

- Pre-requisite: Check the status of any redundant links to ensure that they are functioning properly.

- Pre-requisite: Verify that the network is stable and that no other nodes are experiencing similar issues.

## Potential Root Causes

| Cause | Probability |
|---|---|
| SFP/XFP transceiver degradation | High |
| Fiber cut | Medium |
| Rx/Tx power loss | Low |

## Diagnostic Commands & Expected Outputs

```
Command: show interfaces transceiver BTM-ACC-19/1 detail, Expected Output: Transceiver Information for BTM-ACC-19/1:, Description: This command will display detailed information about the transceiver, including its type, status, and optical parameters. This will help to identify any issues with the transceiver itself.

Command: show interfaces transceiver BTM-ACC-19/1 optical, Expected Output: Optical Parameters for BTM-ACC-19/1:, Description: This command will display the optical parameters for the transceiver, including the received and transmitted power levels. This will help to
```

```
identify any issues with the optical transport.

Command: show interfaces BTM-ACC-19/1 detail, Expected Output:
Interface BTM-ACC-19/1 is up, line protocol is up, Description: This
command will display detailed information about the interface,
including its status and any error counters. This will help to
identify any issues with the interface itself.
```

## Resolution Steps

Step 1: Step: Replace the SFP/XFP transceiver, Command: replace-transceiver BTM-ACC-19/1, Description: This step involves replacing the SFP/XFP transceiver with a new one. The replacement process should be performed according to the vendor's instructions.

Step 2: Step: Repair or replace the fiber optic cable, Command: repair-fiber BTM-ACC-19/1, Description: This step involves repairing or replacing the fiber optic cable. The repair or replacement process should be performed according to the vendor's instructions.

Step 3: Step: Adjust the Rx/Tx power levels, Command: adjust-power BTM-ACC-19/1, Description: This step involves adjusting the Rx/Tx power levels to ensure that they are within the recommended range. The adjustment process should be performed according to the vendor's instructions.

## Post-Check Verification

```
Command: show interfaces transceiver BTM-ACC-19/1 detail, Expected
Output: Transceiver Information for BTM-ACC-19/1: OK, Description:
This command will verify that the transceiver is functioning properly.

Command: show interfaces transceiver BTM-ACC-19/1 optical, Expected
Output: Optical Parameters for BTM-ACC-19/1: OK, Description: This
command will verify that the optical transport is functioning
properly.

Command: show interfaces BTM-ACC-19/1 detail, Expected Output:
Interface BTM-ACC-19/1 is up, line protocol is up, Description: This
command will verify that the interface is functioning properly.
```

## Roll-back Procedures

- Step: Restore the original SFP/XFP transceiver, Command: restore-transceiver BTM-ACC-19/1, Description: This step involves restoring the original SFP/XFP transceiver. The restoration process should be performed according to the vendor's instructions.

- Step: Revert the fiber optic cable repair or replacement, Command: revert-fiber BTM-ACC-19/1, Description: This step involves reverting the fiber optic cable repair or replacement. The reverting process should be performed according to the vendor's instructions.

- Step: Revert the Rx/Tx power level adjustments, Command: revert-power BTM-ACC-19/1, Description: This step involves reverting the Rx/Tx power level adjustments. The reverting process should be performed according to the vendor's instructions.